MAE 143B HW9

6.16 Determine the range of K for which the closed-loop systems (see Fig. 6.18) are stable for each of the cases below by making a Bode plot for $K = 1$ and imagining the magnitude plot sliding up or down until instability results. Verify your answers by using a very rough sketch of a root-locus plot.

(a) $KG(s) = \dfrac{K(s+2)}{s+20}$

(b) $KG(s) = \dfrac{K}{(s+10)(s+1)^2}$

(c) $KG(s) = \dfrac{K(s+10)(s+1)}{(s+100)(s+5)^3}$

6.17 Determine the range of K for which each of the listed systems is stable by making a Bode plot for $K = 1$ and imagining the magnitude plot sliding up or down until instability results. Verify your answers by using a very rough sketch of a root-locus plot.

(a) $KG(s) = \dfrac{K(s+1)}{s(s+5)}$

(b) $KG(s) = \dfrac{K(s+1)}{s^2(s+10)}$

(c) $KG(s) = \dfrac{K}{(s+2)(s^2+9)}$

(d) $KG(s) = \dfrac{K(s+1)^2}{s^3(s+10)}$

6.21 Draw the Nyquist plot for the system in Fig. 6.89. Using the Nyquist stability criterion, determine the range of K for which the system is stable. Consider both positive and negative values of K.

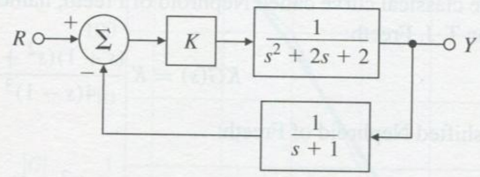

Figure 6.89
Control system for Problem 6.21